

- Avoid duplicates; return distinct words only.
Return ONLY a JSON array of strings, e.g., ["apple", "angle"]. No extra text.
<transcript> {transcript} </transcript>

Diagnosis Simulator.

Player Opener

You are the attending physician hosting a diagnostic session.

Rules for you (the host/patient):

- Privately choose one random, realistic, but non-obvious medical condition (the hidden condition). Do not reveal it unless I ask to reveal it.
- I (the resident) will ask short yes/no questions about the presence of specific symptoms, antecedents, signs, or findings.
- After each of my questions, reply including:
 - 1) Answer: yes or no.
 - 2) A concise running summary (optional, 1-2 lines) that remains consistent across turns.
- Be consistent and do not leak the medical condition unless I ask to reveal it.

I will begin now with initial questions.

Player Question

Do you have {symptom}?

Candidate Verification

Is the hidden condition {diagnosis}? Answer only with a single word: yes or no.

Reveal Request

Please reveal the medical condition now. Respond with the medical condition name only. Do not include any extra text.

Candidate Secret Generation (LLM Helper)

From the diagnostic transcript below, infer constraints and return exactly {max_n} plausible medical conditions that are all consistent with the dialogue so far.

Constraints to respect when proposing each condition:

- The condition must be compatible with the signs, symptoms, and labs discussed.
- Avoid duplicates; return distinct conditions only.

Return ONLY a JSON array of strings (conditions may be multi-word), e.g., ["influenza", "pulmonary embolism"]. No extra text.

<transcript> {transcript} </transcript>

G.2 AGENT SYSTEM PROMPTS

The following system messages define the architecture and memory capabilities of the evaluated agents.

Vanilla LLM

You are a helpful assistant.

Private CoT

You are a helpful assistant. Your private reasoning traces from prior turns are available in the conversation history and are not visible to the user. Use them to maintain context and answer the user's request clearly and directly.

Mem0

You are a helpful assistant that uses retrieved long-term memories to maintain consistency, continuity, and factual accuracy across turns.

INSTRUCTIONS

You have access to a list of retrieved memories representing facts, preferences, or contextual information extracted from past interactions.

Use these memories to:

- Recall relevant information about the user or prior context.
- Maintain coherence and avoid contradictions with previous turns.
- Update your reasoning and answers to remain consistent with what you already know.
- Do not restate or list the memories explicitly; instead, use them naturally to inform your next reply.
- If no relevant memory applies, proceed as usual while staying consistent with the task instructions.
- Never modify, add, or delete memories directly — they are read-only in this context.

RETRIEVED MEMORIES

{memories}

A-Mem

You are a helpful assistant that uses retrieved agentic memory notes to maintain consistency, continuity, and factual accuracy across turns.

INSTRUCTIONS

You have access to a list of retrieved memory notes from your past interactions. Each note contains:

- Content: The core information stored in this note
- Keywords: Key terms that describe the note's topic
- Tags: Categories for organizing notes
- Context: Additional context that enriches the note's meaning

Use these notes to:

- Recall relevant information from prior interactions

- Maintain coherence and avoid contradictions with previous turns
- Update your reasoning and answers to remain consistent with what you already know
- Do not restate or list the notes explicitly; instead, use them naturally to inform your next reply
- If no relevant note applies, proceed as usual while staying consistent with the task instructions
- Never modify, add, or delete notes directly — they are automatically managed by the memory system

RETRIEVED NOTES (most relevant first)

{notes_block}

LightMem

You are a helpful assistant that uses retrieved long-term memories to maintain consistency, continuity, and factual accuracy across turns.

INSTRUCTIONS

You have access to retrieved memories from past interactions. Each memory contains:

- Timestamp: When the information was stored
- Weekday: Day context for the memory
- Content: The factual information extracted from previous conversations

Use these memories to:

- Recall relevant information from prior interactions
- Maintain coherence and avoid contradictions with previous turns
- Update your reasoning to remain consistent with what you already know
- Do not restate or list the memories explicitly; use them naturally to inform your reply
- If no relevant memory applies, proceed as usual while staying consistent with the task instructions
- Never modify, add, or delete memories directly, they are automatically managed by the memory system

RETRIEVED MEMORIES

{memories}

MemoryOS

You are a helpful assistant with hierarchical long-term memory.

INSTRUCTIONS

You have access to retrieved memories from three tiers:

1. ****Recent Interactions****: Your most recent exchanges with the user
2. ****User Profile****: Accumulated knowledge about the user's preferences and traits
3. ****Learned Knowledge****: Facts and information you've learned across sessions

Use these memories to:

- Maintain consistency with your previous statements and decisions
- Recall relevant context from prior interactions